

Capstone-Project

Objective:

The main goal of this project is to provide engineers with hands-on experience in implementing a full DevOps lifecycle in a real-world Azure cloud environment. The project emphasizes DevOps best practices such as automation, collaboration, continuous delivery, security, and observability.

Key Components & Goals:

- **DevOps Culture:** Promote collaboration, automation, continuous improvement.
 - **Source Control:** Manage code using Git and Azure Repos with GitFlow and branch policies.
 - **Infrastructure as Code (IaC):** Provision resources like AKS, ACR, VNets using Terraform.
 - **CI/CD Pipelines:** Automate build, test, and deploy using Azure Pipelines and GitHub Actions.
 - **Security:** Integrate Azure Key Vault, NSGs, Azure Defender, and policy-based compliance.
 - **Monitoring & Logging:** Use Azure Monitor, App Insights for observability and health tracking.
 - **Disaster Recovery (DR):** Implement Active-Passive AKS clusters across regions with failover strategy.
 - **SRE Practices:** Define and track SLA, SLO, and SLI, and simulate failover scenarios.
-

Project Phases Breakdown:

Phase 1: Planning and Version Control

- Establish DevOps culture and KPIs.
- Understand SRE practices (SLA/SLO/SLI).
- Implement Git and GitFlow in Azure Repos.
- Use Azure Boards, Repos, Pipelines, and Artifacts.

Phase 2: Infrastructure as Code (Terraform)

- Provision AKS, ACR, VNets, and secure connectivity (VPN, Bastion, NAT).
- Use Azure Pipelines to automate Terraform deployments.
- VNet peering and simulated on-prem environment with backup.

Phase 3: CI/CD Implementation

- Build CI/CD pipelines in YAML for app and infra.

- Containerize apps, push to ACR, and deploy to AKS clusters.
- Integrate testing tools, Key Vault, and GitHub Actions.
- Create AKS clusters in DR configuration (Active-Passive).

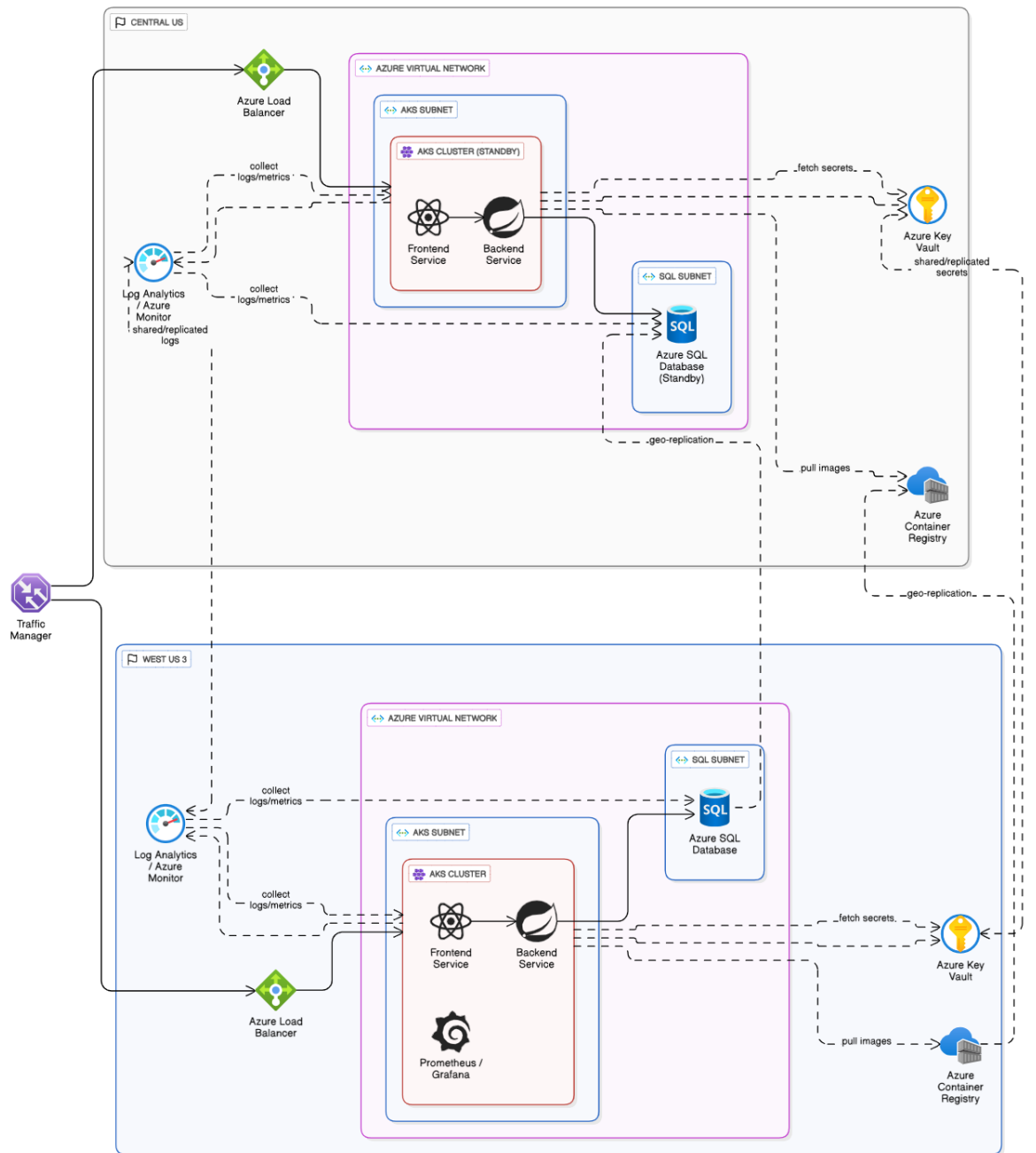
Phase 4: Monitoring, Logging, and Compliance

- Set up Azure Monitor and Application Insights.
- Implement dashboards, alerts, auto-scaling, crash analytics.
- Monitor SLA/SLO compliance and system health.

Phase 5: Security and Governance

- Use Azure Defender for container and registry security.
- Configure NSGs, just-in-time access, Azure Policy.
- Centralized secrets management using Azure Key Vault.

Architecture



I have created 3 pipeline one for infra one for backend one for frontend.

Pipelines

[Recent](#) [All](#) [Runs](#)

[New pipeline](#) ⋮

Filter pipelines

Recently run pipelines

Pipeline	Last run
 frontend	#20250623.7 • Updated react-event-deploy.yaml Manually triggered for  main Yesterday 2m 59s
 backend	#20250623.7 • Updated react-event-deploy.yaml Manually triggered for  main Yesterday 6m 24s
 infra	#20250623.2 • Updated variables.tf Individual CI for  main Yesterday 7m 27s

Capstone-Project / infra /

Add file



yogeshpri Create azure-pipelines.yml

7402b49 · 6 hours ago

History

Name	Last commit message	Last commit date
..		
azure-pipelines.yml	Create azure-pipelines.yml	6 hours ago
main.tf	Create main.tf	9 hours ago
output.tf	Create output.tf	8 hours ago
variables.tf	Create variables.tf	8 hours ago

Form **infra** pipeline demonstrates how to provision a secure, multi-region Kubernetes infrastructure on Azure using **Terraform**, and deploy applications using **Azure DevOps Pipelines**. It includes:

- Automated provisioning of resources using Terraform.
- Active-Passive AKS setup across **Central US** and **West US 3**.
- Azure Traffic Manager for traffic distribution.
- Secure networking with NSGs, Key Vault, and minimal exposure.
- CI/CD pipeline to deploy infrastructure from Azure DevOps.

Terraform Infrastructure Summary

Resources Created:

- **Resource Group:** yogesh-caps-rg
- **Virtual Networks & Subnets:**
 - VNet with public & private subnets per region
- **NSGs:**
 - Allow HTTP (80), HTTPS (443), and custom ports (8080, 8081)
- **AKS Clusters:**
 - Two clusters in centralus and westus3 using private subnets
- **Public IPs:**
 - Static public IPs per AKS cluster for ingress
- **Traffic Manager:**
 - Routes traffic to AKS clusters based on performance
- **Container Registry:**
 - ACR for storing container images
- **Key Vault:**
 - Stores secrets, with access granted to AKS MSI
- **Output Variables:**
 - Public IPs, kubeconfigs, Key Vault URI, ACR credentials

6. Security Controls

- **NSGs:** Only required ports are open.
- **Key Vault:** Access policy restricts secret access to AKS only.
- **Private subnets:** AKS nodes are deployed in isolated subnets.
- **ACR:** Admin access enabled (can be switched to MSI later for security).

After applying it created all the resources

Create

Manage view

Delete resource group

Refresh

Export to CSV

Open query

Assign tags

Move

Filter for any field...

Type equals all

Location equals all

Add filter

Showing 1 to 37 of 37 records.

Show hidden types

No grouping

List view

Name	Type	Location
aks-1	Kubernetes service	Central US
aks-2	Kubernetes service	West US 3
KubernetesRecordingRulesRuleGroup - aks-1	Prometheus rule group	West US 3
KubernetesRecordingRulesRuleGroup - aks-2	Prometheus rule group	West US 3
KubernetesRecordingRulesRuleGroup - aks-active-cluster	Prometheus rule group	Central US
MSCI-eastus-aks-1	Data collection rule	East US
MSCI-eastus-aks-2	Data collection rule	East US

< Previous

Page 1 of 1

Next >

Give feedback

Backend:

Stage: Build_Backend

- **Tool Used:** Docker@2
- **Action:** Builds and pushes the Java backend image to **Azure Container Registry (ACR)**.
- **Key Inputs:**
 - Dockerfile path: Backend-Java/Event_Managment/Dockerfile
 - Image: yogcregistry.azurecr.io/backend/event-management:\$(Build.BuildId)

Jobs in run #202...
backend

Build and Push Backend Image

BuildBackend20s

Initialize job<1s

Checkout Caps...3s

Build and pus...14s

Post-job: Che...<1s

Finalize Job<1s

Run Trivy Scan on Image

Scan Docker I...1m 8s

Build and push Backend image

240 createdAt:2025-06-17T09:17:40Z; layerSize:0B; createdBy:WORKDIR /app; layerId:<miss

241 createdAt:2025-04-23T14:48:05Z; layerSize:0B; createdBy:ENTRYPOINT ["/__cacent_entr

242 createdAt:2025-04-23T14:48:05Z; layerSize:5.31kB; createdBy:COPY --chmod=755 entryyp

243 createdAt:2025-04-23T14:48:05Z; layerSize:0B; createdBy:RUN /bin/sh -c set -eux;

244 createdAt:2025-04-23T14:48:05Z; layerSize:164MB; createdBy:RUN /bin/sh -c set -eux;

245 createdAt:2025-04-23T14:48:05Z; layerSize:0B; createdBy:ENV JAVA_VERSION=jdk-21.0.7

246 createdAt:2025-04-23T14:48:05Z; layerSize:32.9MB; createdBy:RUN /bin/sh -c set -eux

247 createdAt:2025-04-23T14:48:05Z; layerSize:0B; createdBy:ENV LANG=en_US.UTF-8 LANGUA

248 createdAt:2025-04-23T14:48:05Z; layerSize:0B; createdBy:ENV PATH=/opt/java/openjdk/

249 createdAt:2025-04-23T14:48:05Z; layerSize:0B; createdBy:ENV JAVA_HOME=/opt/java/ope

250 createdAt:2025-02-14T03:28:36Z; layerSize:0B; createdBy:CMD ["/bin/sh"]; layerId:<m

251 createdAt:2025-02-14T03:28:36Z; layerSize:7.83MB; createdBy:ADD alpine-minirootfs-3

252

253 /usr/bin/docker inspect 6eab13b96e801225fdb367f0dabe098be4150b53b324c5abf521f01f402

254 [sha256:08000c18d16dadf9553d747a58cf44023423a9ab010aab96cf263d2216b8b350 sha256:343

255

256 Finishing: Build and push Backend image

Stage: TrivyScan

- **Tool Used:** Trivy CLI
- **Action:** Scans the backend Docker image for **vulnerabilities** (HIGH, CRITICAL, MEDIUM).
- **Fail Criteria:** Build fails if vulnerabilities are detected and not ignored.

Jobs in run #20250623.7

Initialize job<1s

Checkout Caps...4s

Install Trivy3s

Run Trivy Scan<1s

Maven58s

Post-job: Che...<1s

Finalize Job<1s

Deploy to AKS1

Deploy Backend...13s

✓Maven

174 Uploading 61 files
175 Using default max parallelism.
176 Creating dedup client:
177 - Max dedup parallelism: 192
178 - Using blobstore domain: 0
179 - Using redirect timeout:
180 - Hash type: Dedup64K
181 Building file tree
182 Uploaded 0 out of 1,074,287 bytes.
183 Uploaded 1,042,979 out of 1,074,287 bytes.
184 Associating files
185 Total files: 61 ---- Associated files: 0 (0%)
186 Total files: 61 ---- Associated files: 52 (85%)
187 File upload succeed.
188 Published '/home/yogesh/_work/6/s/Backend-Java/Event_Managment/target/site/jacoco' a
189 Async Command End: Publish code coverage
190 Finishing: Maven

2025-06-23T09:58:31.1558332Z
2025-06-23T09:58:31.1578379Z Report Summary
2025-06-23T09:58:31.1578777Z
2025-06-23T09:58:31.1899401Z
2025-06-23T09:58:31.1903183Z
2025-06-23T09:58:31.1904223Z
2025-06-23T09:58:31.1905001Z
2025-06-23T09:58:31.1905649Z
2025-06-23T09:58:31.1906656Z
2025-06-23T09:58:31.1907936Z
2025-06-23T09:58:31.1909041Z
2025-06-23T09:58:31.1909448Z
2025-06-23T09:58:31.1909908Z
2025-06-23T09:58:31.1910264Z
2025-06-23T09:58:31.1910424Z
2025-06-23T09:58:31.1910525Z
2025-06-23T09:58:31.1910790Z
2025-06-23T09:58:31.1911236Z
2025-06-23T09:58:31.1911727Z
2025-06-23T09:58:31.1911894Z
2025-06-23T09:58:31.1912943Z

Target	Type	Vulnerabilities	Secrets
yogeshacr1.azurecr.io/yogeshacr1.azurecr.io/backend/event-management:58 (alpine 3.21.3)	alpine	0	-
app/app.jar	jar	13	-

Legend:
- '-': Not scanned
- '0': Clean (no security findings detected)
2025-06-23T09:58:31.1910790Z Java (jar)
2025-06-23T09:58:31.1911236Z =====
2025-06-23T09:58:31.1911727Z Total: 13 (MEDIUM: 5, HIGH: 7, CRITICAL: 1)
2025-06-23T09:58:31.1911894Z
2025-06-23T09:58:31.1912943Z

Title	Library	Vulnerability	Severity	Status	Installed Version	Fixed Version
ch.qos.logback:logback-core (app.jar)		CVE-2024-12798	MEDIUM	fixed	1.5.12	1.5.13, 1.3.15
logback-core: arbitrary code execution via						
JaninoEventEvaluator						
https://avd.aquasec.com/nvd/cve-2024-12798						

```
2025-06-23T09:58:31.2012945Z | CVE-2025-49125 | | | | 11.0.8, 10.1.42, 9.0.106
| tomcat: Apache Tomcat: Security constraint bypass for
2025-06-23T09:58:31.2013798Z | | | | |
| pre/post-resources
2025-06-23T09:58:31.2014895Z | | | | |
| https://avd.aquasec.com/nvd/cve-2025-49125
2025-06-23T09:58:31.2015961Z | | | | |
2025-06-23T09:58:31.2017161Z | org.springframework.boot:spring-boot (app.jar) | CVE-2025-22235 | HIGH | 3.4.1 | 3.3.11, 3.4.5
| org.springframework.boot:spring-boot: Spring Boot
2025-06-23T09:58:31.2034181Z | | | | |
| EndpointRequest.to() creates wrong matcher if actuator
2025-06-23T09:58:31.2035280Z | | | | |
| endpoint is not...
2025-06-23T09:58:31.2036126Z | | | | |
| https://avd.aquasec.com/nvd/cve-2025-22235
2025-06-23T09:58:31.2041419Z | | | | |
2025-06-23T09:58:31.2042763Z | org.springframework:spring-web (app.jar) | CVE-2025-41234 | MEDIUM | 6.2.1 | 6.2.8, 6.1.21
| springframework: Reflected download attack in Spring
2025-06-23T09:58:31.2044087Z | | | | |
| Framework with non-ASCII headers
2025-06-23T09:58:31.2045292Z | | | | |
| https://avd.aquasec.com/nvd/cve-2025-41234
2025-06-23T09:58:31.2046636Z | | | | |
2025-06-23T09:58:31.2046923Z
2025-06-23T09:58:31.2062731Z ##[section]Finishing: Run Trivy Scan
```

Maven Build & Unit Testing

- Tool Used: Maven@4
- Action:
 - Builds the Java project.
 - Runs unit tests using Surefire.
 - Generates JaCoCo test coverage reports.
 - Publishes JUnit results for visibility in Azure DevOps.

✓ #20250623.7 • Updated react-event-deploy.yaml

📁 backend

Run new

🕒 This run is being retained as one of 3 recent runs by main (Branch).

View retention leases

SummaryTestsExtensionsCode Coverage

Event_Managment

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed Cxty	Missed Lines	Missed Methods	Missed Classes				
com.EventManagment.service		3%		0%	31	34	47	50	30	33	3	6
com.EventManagment.Controllers		7%		n/a	14	17	24	27	14	17	1	4
com.EventManagment.model		0%		0%	2	2	1	1	1	1	1	1
com.EventManagment		37%		n/a	1	2	2	3	1	2	0	1
com.EventManagment.Config		100%		n/a	0	2	0	5	0	2	0	1
Total	370 of 429	13%	4 of 4	0%	48	57	74	86	46	55	5	13

Created with [JaCoCo](#) 0.8.11.202310140853

Stage: Security_Scan (Snyk)

- Tool Used: Snyk CLI
- Action:

- Authenticates using SNYK_TOKEN
- Performs deep **open-source dependency scanning** across all projects.
- Outputs results in JSON format for compliance tracking.

← Jobs in run #20250623.7

✓ Initialize job <1s

✓ Checkout Caps... 3s

✓ Ensure Node.... <1s

✓ Install Snky CLI 8s

✓ Authenticate S... 3s

✓ Run Snky Test 7s

✓ Post-job: Che... <1s

✓ Finalize Job <1s

SonarCloud Code Analysis

> ✓ Analyze Jav... 1m 28s

✓ Authenticate Snky

1 Starting: Authenticate Snky

2 =====

3 Task : Command line

4 Description : Run a command line script using Bash on Linux and macOS and cmd.exe on Window

5 Version : 2.250.1

6 Author : Microsoft Corporation

7 Help : <https://docs.microsoft.com/azure/devops/pipelines/tasks/utility/command-line>

8 =====

9 Generating script.

10 ===== Starting Command Output =====

11 /usr/bin/bash --noprofile --norc /home/yogesh/_work/_temp/b8a97213-8612-4aba-a8f4-cd1821aff7

12 Authenticating with Snky...

13

14 Your account has been authenticated. Snky is now ready to be used.

15

16

17 Finishing: Authenticate Snky

View raw log

```

2025-06-23T10:01:37.3758411Z },
2025-06-23T10:01:37.3758664Z "upgrade": {
2025-06-23T10:01:37.3759087Z   "axios@1.7.9": {
2025-06-23T10:01:37.3759675Z     "upgradeTo": "axios@1.8.3",
2025-06-23T10:01:37.3759933Z     "upgrades": [
2025-06-23T10:01:37.3760147Z       "axios@1.7.9",
2025-06-23T10:01:37.3760388Z       "axios@1.7.9"
2025-06-23T10:01:37.3760610Z     ],
2025-06-23T10:01:37.3760818Z     "vulns": [
2025-06-23T10:01:37.3761044Z       "SNYK-JS-AXIOS-9403194",
2025-06-23T10:01:37.3761274Z       "SNYK-JS-AXIOS-9292519"
2025-06-23T10:01:37.3761502Z     ]
2025-06-23T10:01:37.3761727Z   },
2025-06-23T10:01:37.3761933Z },
2025-06-23T10:01:37.3762127Z "patch": {},
2025-06-23T10:01:37.3762352Z "ignore": {},
2025-06-23T10:01:37.3762568Z "pin": {}
2025-06-23T10:01:37.3762784Z },
2025-06-23T10:01:37.3763009Z "filesystemPolicy": false,
2025-06-23T10:01:37.3763228Z "filtered": {
2025-06-23T10:01:37.3763462Z   "ignore": [],
2025-06-23T10:01:37.3763694Z   "patch": []
2025-06-23T10:01:37.3763927Z },
2025-06-23T10:01:37.3764146Z "uniqueCount": 12,
2025-06-23T10:01:37.3764425Z "projectName": "event-management-dashboard",
2025-06-23T10:01:37.3764710Z "foundProjectCount": 1,
2025-06-23T10:01:37.3765045Z "displayTargetFile": "Frontend-ReactJs/event-management-dashboard/package-lock.json",
2025-06-23T10:01:37.3765557Z "hasUnknownVersions": false,
2025-06-23T10:01:37.3765991Z "path": "/home/yogesh/_work/6/s"
2025-06-23T10:01:37.3766226Z }
2025-06-23T10:01:37.3766327Z
2025-06-23T10:01:37.3772817Z ##[section]Finishing: Run Snky Test

```

Stage: SonarCloudAnalysis

- **Tools Used:** SonarCloud, Maven
- **Action:**
 - Prepares and runs SonarCloud static code analysis.

- Publishes a **Quality Gate** with metrics such as code smells, bugs, coverage, and maintainability.

Summary Tests Extensions Code Coverage

SonarQube Cloud Analysis Report

 **Quality Gate passed (newprac)**

Issues

 0 New issues

 0 Accepted issues

Measures

 0 Security Hotspots

[See analysis details on SonarQube Cloud](#)

Summary Tests Extensions Code Coverage

Summary

2 Run(s) Completed (2 Passed, 0 Failed)

Total tests: 2 (2 Passed, 0 Failed, 0 Others)

Pass percentage: 100%

Run duration: 27s 873ms

Tests not reported: 0

 Bug ▾

 Link

 Associate Test Case

 Test run ▾

 Filter by test or run name

Tags ▾

Test file ▾

Owner

← Jobs in run #20250623.7

✓ Analyze Jav... 1m 28s

✓ Initialize job <1s

✓ Checkout Caps... 2s

✓ Prepare Sonar... 4s

✓ Run Mav... 1m 13s

✓ Publish Sonar... 5s

✓ Post-job: Che... <1s

✓ Finalize Job <1s

Finalize build

✓ Publish SonarCloud Quality Gate

View raw log

```
1 Starting: Publish SonarCloud Quality Gate
2 =====
3 Task      : Publish Quality Gate Result
4 Description : Publish SonarQube Cloud's Quality Gate result on the Azure Pipelines build result
5 Version    : 3.3.0
6 Author     : sonarsource
7 Help       : [More Information](https://docs.sonarcloud.io/advanced-setup/ci-based-analysis/s
8 =====
9 [INFO] SonarQube Cloud: Server version: 8.0.0.64674
10 [INFO] SonarQube Cloud: Task AZecPihIJewhxeK_-uks completed
11 [INFO] SonarQube Cloud: Analysis succeeded with warning: The property 'sonar.login' is deprec
12 [INFO] SonarQube Cloud: Analysis succeeded with warning: Shallow clone detected during the ana
13 [INFO] SonarQube Cloud: Overall Quality Gate status: ok
14 Finishing: Publish SonarCloud Quality Gate
```

← Jobs in run #20250623.7

✓ Analyze Jav... 1m 28s

✓ Initialize job <1s

✓ Checkout Caps... 2s

✓ Prepare Sonar... 4s

✓ Run Mav... 1m 13s

✓ Publish Sonar... 5s

✓ Post-job: Che... <1s

✓ Finalize Job <1s

Finalize build

✓ Run Maven Build & SonarCloud Analysis

View raw log

```
342 [INFO] 10:03:30.195 Analysis total time: 34.314 s
343 [INFO] -----
344 [INFO] BUILD SUCCESS
345 [INFO] -----
346 [INFO] Total time: 01:00 min
347 [INFO] Finished at: 2025-06-23T10:03:30Z
348 [INFO] -----
349
350 Result Attachments will be stored in LogStore
351 Run Attachments will be stored in LogStore
352 Async Command Start: Publish test results
353 Publishing test results to test run '23'.
354 TestResults To Publish 1, Test run id:23
355 Test results publishing 1, remaining: 0. Test run id: 23
356 Published Test Run : https://dev.azure.com/devuser110591/caps/_TestManagement/Runs?runId
357 Async Command End: Publish test results
358 Finishing: Run Maven Build & SonarCloud Analysis
```

Search for projects

Perspective

Overall Status

Sort by Name

1 projects

★ devuser110591 / newprac New Private

✓ Passed

Last analysis: 6/23/2025, 3:32 PM • 390 Lines of Code • Java

A 0 Security

A 9 Reliability

A 35 Maintainability

E 0.0% Hotspots Reviewed

0.0% Coverage

0.0% Duplications

1 of 1 shown

Stage: DeployToAKS1

- **Target Cluster:** aks-1 (Primary - Central US)
- **Tool Used:** Kubernetes@1
- **Manifests Applied:**
 - springboot-deploy.yaml
 - java-springboot-service.yaml
- **Purpose:** Deploy backend (and frontend optionally) to primary AKS.

The screenshot displays an Azure DevOps pipeline interface. On the left, a list of jobs for run #20250623.7 is shown. The 'Deploy Backend...' job is expanded, showing its steps: 'Initialize job', 'Checkout Caps...', 'Apply Kubernetes Manifests' (highlighted), 'Post-job: Check...', and 'Finalize Job'. The 'Apply Kubernetes Manifests' step is currently running. On the right, a terminal window titled 'Apply Kubernetes Manifests' shows the execution of a Kubernetes manifest. The manifest defines a LoadBalancer service with an ingress rule pointing to a specific IP address (72.152.47.157) in VIP mode. The terminal output shows the manifest being applied to the 'aks-1' cluster.

```
150         "type": "LoadBalancer"
151     },
152     "status": {
153         "loadBalancer": {
154             "ingress": [
155                 {
156                     "ip": "72.152.47.157",
157                     "ipMode": "VIP"
158                 }
159             ]
160         }
161     }
162 }
163 ]
164 }
165
166 Finishing: Apply Kubernetes Manifests
```

Stage: DeployToAKS2

- **Target Cluster:** aks-2 (Secondary - West US 3)
- **Tool Used:** Kubernetes@1
- **Manifests Applied:**
 - springboot-deploy.yaml
 - java-springboot-service.yaml
- **Purpose:** Deploy same services to the secondary AKS cluster to support **Active-Passive failover** via Azure Traffic Manager.

←

Jobs in run #20250623.7

▲

✓

Post-job: Che...

<1s

✓

Finalize Job

<1s

Deploy to AKS2

▼

✓

Deploy Backend ...

7s

✓

Initialize job

<1s

✓

Checkout Caps...

3s

✓

Apply Kuberne...

3s

✓

Post-job: Che...

<1s

✓

Finalize Job

<1s

security scan with snyk

✓

Apply Kubernetes Manifests

Q

View

150 "\"type\": \"LoadBalancer\"
151 },
152 \"status\": {
153 \"loadBalancer\": {
154 \"ingress\": [
155 {
156 \"ip\": \"20.14.109.66\",
157 \"ipMode\": \"VIP\"
158 }
159]
160 }
161 }
162 }
163]
164 }
165 }
166 Finishing: Apply Kubernetes Manifests

Frontend:

Stage: Build_Frontend

- Purpose: Build the frontend React app and push the container image to Azure Container Registry (ACR).
- Tool: Docker@2 task
- Key Inputs:
 - Dockerfile: Frontend-ReactJs/event-management-dashboard/Dockerfile
 - Image pushed to: yogcregistry.azurecr.io/frontend/event-dashboard:\$(Build.BuildId)

← Jobs in run #202...
frontend

Build and Push Frontend Image

✓ BuildFrontend 11s

✓ Initialize job <1s

✓ Checkout Caps... 2s

✓ Build and push... 6s

✓ Post-job: Che... <1s

✓ Finalize Job <1s

Run Trivy Scan on Image

✓ Scan Docker Ima... 8s

✓ Build and push Frontend image

280 createdAt:2025-04-16T14:50:31Z; layerSize:4.62kB; createdBy:COPY 30

281 createdAt:2025-04-16T14:50:31Z; layerSize:3.02kB; createdBy:COPY 20

282 createdAt:2025-04-16T14:50:31Z; layerSize:389B; createdBy:COPY 15-16

283 createdAt:2025-04-16T14:50:31Z; layerSize:2.12kB; createdBy:COPY 10

284 createdAt:2025-04-16T14:50:31Z; layerSize:1.62kB; createdBy:COPY do

285 createdAt:2025-04-16T14:50:31Z; layerSize:4.05MB; createdBy:RUN /bin

286 createdAt:2025-04-16T14:50:31Z; layerSize:0B; createdBy:ENV DYNPKG_

287 createdAt:2025-04-16T14:50:31Z; layerSize:0B; createdBy:ENV PKG_REL

288 createdAt:2025-04-16T14:50:31Z; layerSize:0B; createdBy:ENV NGINX_V

289 createdAt:2025-04-16T14:50:31Z; layerSize:0B; createdBy:LABEL mainta

290 createdAt:2025-02-14T03:28:36Z; layerSize:0B; createdBy:CMD ["/bin/s

291 createdAt:2025-02-14T03:28:36Z; layerSize:7.83MB; createdBy:ADD alp

292

293 /usr/bin/docker inspect cf497b0351ff29416432c0909dad7a9ebd84f2749094

294 [sha256:08000c18d16dadf9553d747a58cf44023423a9ab010aab96cf263d2216b8

295

296 Finishing: Build and push Frontend image

Stage: TrivyScan

- **Purpose:** Perform container **vulnerability scanning** using **Trivy**.
- **Tool:** Trivy CLI
- **Action:**
 - Install Trivy
 - Scan the built Docker image for vulnerabilities (HIGH, CRITICAL, MEDIUM)
 - Fail the build if severe vulnerabilities are found

```
2025-06-23T10:19:22.3173601Z
2025-06-23T10:19:22.3174574Z Report Summary
2025-06-23T10:19:22.3175198Z
2025-06-23T10:19:22.3597202Z
2025-06-23T10:19:22.3598345Z
2025-06-23T10:19:22.3599083Z
2025-06-23T10:19:22.3599704Z
2025-06-23T10:19:22.3600229Z
2025-06-23T10:19:22.3601007Z
2025-06-23T10:19:22.3601286Z
2025-06-23T10:19:22.3601507Z - '-': Not scanned
2025-06-23T10:19:22.3601789Z - '0': Clean (no security findings detected)
2025-06-23T10:19:22.3601910Z
2025-06-23T10:19:22.3601981Z
2025-06-23T10:19:22.3602280Z yোগেশacr1.azurecr.io/yোগেশacr1.azurecr.io/frontend/event-dashboard:36 (alpine 3.21.3)
2025-06-23T10:19:22.3602692Z =====
2025-06-23T10:19:22.3603196Z Total: 2 (MEDIUM: 0, HIGH: 2, CRITICAL: 0)
2025-06-23T10:19:22.3603317Z
2025-06-23T10:19:22.3604256Z
2025-06-23T10:19:22.3604892Z
2025-06-23T10:19:22.3605701Z
2025-06-23T10:19:22.3607046Z
2025-06-23T10:19:22.3607997Z
2025-06-23T10:19:22.3609241Z
2025-06-23T10:19:22.3609898Z
2025-06-23T10:19:22.3610521Z
2025-06-23T10:19:22.3611250Z
2025-06-23T10:19:22.3611436Z
2025-06-23T10:19:22.3643889Z ##[section]Finishing: Run Trivy Scan
```

Target	Type	Vulnerabilities	Secrets
yোগেশacr1.azurecr.io/yোগেশacr1.azurecr.io/frontend/event-dashboard:36 (alpine 3.21.3)	alpine	2	-

Legend:

2025-06-23T10:19:22.3601507Z - '-': Not scanned

2025-06-23T10:19:22.3601789Z - '0': Clean (no security findings detected)

2025-06-23T10:19:22.3601910Z

2025-06-23T10:19:22.3601981Z

2025-06-23T10:19:22.3602280Z yোগেশacr1.azurecr.io/yোগেশacr1.azurecr.io/frontend/event-dashboard:36 (alpine 3.21.3)

2025-06-23T10:19:22.3602692Z =====

2025-06-23T10:19:22.3603196Z Total: 2 (MEDIUM: 0, HIGH: 2, CRITICAL: 0)

Library	Vulnerability	Severity	Status	Installed Version	Fixed Version	Title
libxml2	CVE-2025-32414	HIGH	fixed	2.13.4-r5	2.13.4-r6	libxml2: Out-of-Bounds Read in libxml2 https://avd.aquasec.com/nvd/cve-2025-32414
	CVE-2025-32415					libxml2: Out-of-bounds Read in xmlSchemaIDCfillNodeTables https://avd.aquasec.com/nvd/cve-2025-32415

Stage: Security_Scan

- **Purpose:** Run a **Snyk** security scan for open-source vulnerabilities.
- **Tools:**
 - Node.js (to run Snyk CLI)
 - Snyk CLI (npm install -g snyk)
- **Action:**
 - Authenticate with Snyk using SNYK_TOKEN
 - Scan all project dependencies for known vulnerabilities
 - Output a JSON report

security scan with snyk

✓	Run Snyk for SCM	24s
✓	Initialize job	<1s
✓	Checkout Caps...	3s
✓	Ensure Node....	<1s
✓	Install Snyk CLI	7s
✓	Authenticate S...	3s
✓	Run Snyk Test	7s
✓	Post-job: Che...	<1s

```
4151     "ignore": {},
4152     "pin": {}
4153   },
4154   "filesystemPolicy": false,
4155   "filtered": {
4156     "ignore": [],
4157     "patch": []
4158   },
4159   "uniqueCount": 12,
4160   "projectName": "event-management-dashboard",
4161   "foundProjectCount": 1,
4162   "displayTargetFile": "Frontend-ReactJs/event-management-dashboard/package-lock.json",
4163   "hasUnknownVersions": false,
4164   "path": "/home/yogesh/_work/5/s"
4165 }
4166
4167 Finishing: Run Snyk Test
```

Stage: DeployToAKS1

- **Cluster:** aks-1 (Primary – Central US)
- **Tool:** Kubernetes@1
- **Manifests Applied:**
 - Kubernetes/react-event-deploy.yaml
 - Kubernetes/react-event-service.yaml
- **Purpose:** Deploy the latest frontend image to the primary AKS cluster

←	Jobs in run #20250623.7	▲
✓	Finalize Job	<1s
Deploy to AKS1		
✓	Deploy Backend ...	8s
✓	Initialize job	<1s
✓	Checkout Caps...	3s
✓	Apply Kuberne...	3s
✓	Post-job: Che...	<1s
✓	Finalize Job	<1s
Deploy to AKS2		
>	Deploy Backend ...	8s

✓ Apply Kubernetes Manifests

```
153     "type": "LoadBalancer"
154   },
155   "status": {
156     "loadBalancer": {
157       "ingress": [
158         {
159           "ip": "4.249.184.164",
160           "ipMode": "VIP"
161         }
162       ]
163     }
164   }
165 }
166 ]
167 }
168
169 Finishing: Apply Kubernetes Manifests
```


Stage: DeployToAKS2

- **Cluster:** aks-2 (Secondary – West US 3)
- **Tool:** Kubernetes@1
- **Manifests Applied:**
 - Kubernetes/react-event-deploy.yaml
 - Kubernetes/react-event-service.yaml
- **Purpose:** Ensure deployment is mirrored to the DR cluster for failover readiness

the impacted pipelines, please follow the [blog](#).

← Jobs in run #20250623.7

✓	Finalize Job	<1s
Deploy to AKS2		
✓	Deploy Backend ...	8s
✓	Initialize job	<1s
✓	Checkout Caps...	2s
✓	Apply Kuberne...	3s
✓	Post-job: Che...	<1s
✓	Finalize Job	<1s
Finalize build		
✓	Report build sta...	<1s

✓ Apply Kubernetes Manifests

```
150         "type": "LoadBalancer"
151     },
152     "status": {
153         "loadBalancer": {
154             "ingress": [
155                 {
156                     "ip": "4.227.40.208",
157                     "ipMode": "VIP"
158                 }
159             ]
160         }
161     }
162 }
163 ]
164 }
165
166 Finishing: Apply Kubernetes Manifests
```

Monitoring

Prometheus-Grafana

Prometheus and Grafana were deployed in a dedicated monitoring namespace in AKS. Prometheus collects metrics from Kubernetes components and application pods, while Grafana provides dashboards to visualize cluster health, resource usage, and application performance. Default and custom dashboards were set up to monitor CPU, memory, and pod states in real time.

